RAFIUL HAQ

Senior Environmental Consultant | Ecologist | Biodiversity & Ecosystem Management | Climate Change & Nature-based Solutions

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PROFESSIONAL SUMMARY

Ecologist and independent environmental consultant with more than three decades of experience across research, conservation programme management and environmental advisory work in Pakistan. Core team member for the development of Pakistan's National Wetlands Policy, former National Coordinator of IUCN's Mangroves for the Future (MFF) programme in Pakistan, and, since 2012, a freelance consultant to the World Bank, Asian Development Bank, WWF-Pakistan, IUCN, SGS, GEF-UNDP Small Grants Programme, Sindh Forest Department and Sindh EPA.

Leads biodiversity components of EIA/ESIA studies for energy, port, water-resource and transport projects; prepares mangrove conservation and climate adaptation plans for coastal Sindh; and provides technical input to blue and green carbon projects, including Delta Blue Carbon (DBC-1) and a pre-feasibility study for Mitsubishi Corporation. Known for translating field-based ecological evidence into practical, policy-aligned and community-focused recommendations, and for extensive training, lecturing and stakeholder facilitation.

CORE COMPETENCIES

Environmental Consulting • Ecology & Biodiversity Assessment • Ecological Baseline Studies
Environmental Impact Assessment (EIA) • Environmental & Social Impact Assessment (ESIA) • ESMF
Wetland Rehabilitation • Mangrove Conservation & Restoration • Coastal Ecosystem Management
Climate Change Adaptation (LAPA) • Nature-based Solutions • Ecosystem Management
Blue & Green Carbon Project Feasibility • Carbon Due Diligence • ESG & Sustainability Advisory
Environmental Policy Development • Natural Resource Management • Compensatory Plantation Planning
Stakeholder Engagement • Programme Management • Technical Leadership • Training & Facilitation

CAREER HIGHLIGHTS

- **National Wetlands Policy (Pakistan):** Member of the core team (2007–2009); coordinated stakeholders and led policy dialogue at sub-national and national levels.
- **Delta Blue Carbon (DBC-1):** As Country Director, Programs and Operations, Indus Delta Capital, facilitated the programme in achieving validation status through a Verra-accredited body (2021); later provided technical due diligence for a carbon-credit buyer (2021–2022).
- **Mangrove conservation, coastal Sindh:** Led cluster-based mangrove conservation plans and sectoral LAPAs for agriculture and fisheries in Ketibundar and Kharochaan for WWF-Pakistan.
- **Biodiversity lead on EIA/ESIA:** Led biodiversity components of assessments for World Bank, ADB, SGS, GEMS and other clients across oil & gas, power, ports, dams and irrigation.
- **Mangroves for the Future (MFF) – Pakistan:** National Coordinator; developed the National Strategy and Action Plan and Small Grants Facility guidelines; convened seven National Coordinating Body meetings with 85–90% member attendance.
- **Advisory roles:** Member, advisory group for DG Parks & Horticulture, KMC; advisory committee for Sindh EPA on technical issues; core group of the Urban Forest Coalition.

PROFESSIONAL EXPERIENCE

Independent Environmental Consultant (Freelance) — Aug 2012 – Present

Founder Member, CRAB (Conservation Restoration Alliance for Biodiversity) Green Solutions — Karachi, Pakistan

- Consultant/advisor to Sindh Forest Department, SGS Environment Division, GEF-UNDP Small Grants Programme Pakistan, The World Bank, Asian Development Bank, LEAD Pakistan, IUCN and IUCN Commission on Ecosystem Management, WWF-Pakistan, Hashoo Foundation, SECMC, Sindh EPA and the Directorate of Parks & Horticulture, KMC.
- Lead biodiversity specialist on EIA/ESIA and ESMF studies, assessing biodiversity sensitivities and risks and recommending mitigation measures that are shared with stakeholders.
- Provide ESG and biodiversity advisory grounded in field evidence, supporting IFC and EBRD biodiversity and ecosystem safeguard requirements.

- Develop mangrove conservation, climate adaptation and compensatory plantation plans; advise Sindh EPA and Parks & Horticulture on biodiversity-linked urban greening.

Selected Consulting Assignments

Pre-feasibility: Green Carbon Sequestration Project — *Mitsubishi Corporation (through IUCN Pakistan)* | *Lead Expert* | *Dec 2022 – Feb 2023 (planned)*
Analysed parameters of the riverine forestry climate-change project, evaluated business viability of carbon project propositions from Sindh Forest Department and provided recommendations.

Ecological Baseline and Vulnerability Assessment — *WWF-Pakistan (through NEC Consultants)* | *Team Lead* | *Jul 2022 – Dec 2022*
Collected ecological baseline and vulnerability information from project villages to set baselines against project indicators.

Due Diligence of Delta Blue Carbon-I — *Carbon Growth (Australian consortium)* | *Technical Expert* | *Nov 2021 – Sep 2022*
Contributed the technical aspects of due diligence for DBC-1 on behalf of the buyer of carbon credits.

Delta Blue Carbon-I — *Indus Delta Capital* | *Country Director, Programs and Operations* | *Apr 2021 – Sep 2021*
Facilitated the programme to achieve validation status from a Verra-accredited agency, establishing eligibility for carbon credits from Indus Delta mangrove forests.

Cluster-based Mangrove Conservation Plan, Ketibundar and Kharochaan — *WWF-Pakistan* | *Team Lead* | *Apr 2021 – Dec 2021*
Developed cluster-based mangrove conservation plans of action (17 villages) reflecting local needs and climate change considerations.

Mangrove Plantation on 25 Acres near Port Qasim — *DP World* | *Team Lead* | *Jun 2021 – Sep 2021*
CSR initiative implemented under CRAB in collaboration with the Coastal Forestry Division, Sindh Forest Department.

Sectoral LAPA for Agriculture and Fisheries, Ketibundar and Kharochaan — *WWF-Pakistan* | *Team Lead* | *Dec 2020 – Jan 2021*
Developed sectoral Local Adaptation Plans of Action reflecting the specifics of climate change for agriculture and fisheries.

Plantation Plan for BRT Green Line Extension, Karachi — *SIDCL (through EMC Consultants)* | *Lead Ecologist* | *Apr 2020 – Sep 2020*
Documented and assessed damage to existing vegetation and proposed an alternative compensatory plantation plan.

Damage Assessment of Mangroves in Port Qasim Area — *Lucky Electric Power Company Limited* | *Team Lead* | *Jan 2019 – Apr 2020*
Designed and implemented an assessment programme across a 1.53 km strip of mangrove habitat.

Berth 3 & 4 Plantation Plan — *Berth 3 & 4 Coal Storage (through GEMS)* | *Lead Plantation Expert* | *Mar 2019 – Sep 2019*
Developed and implemented the plantation plan.

ESIA & ESMF of Ecotourism Projects in Khyber Pakhtunkhwa — *The World Bank* | *Reviewer, Biodiversity Section* | *Jan 2019 – Feb 2019*
Reviewed and contributed to the biodiversity section of the document.

Peshawar–Torkham Economic Corridor ESIA — *The World Bank* | *Lead Biodiversity* | *Mar 2018 – Apr 2018*
Contributed the biodiversity section of the ESIA.

ESIA for Development of Tanda Dam as an Ecotourism Resort — *DAI Global LLC (through NEC Consultants)* | *Lead Biodiversity* | *Oct 2017 – Feb 2018*
Contributed the biodiversity section of the ESIA.

Balochistan Water Resource Development Project (ADB TA 8800-PAK) — *Asian Development Bank (through Techno Consult International)* | *Lead Biodiversity* | *May 2017 – Nov 2017*
Biodiversity input for a proposed 70 m earth-fill dam on the Sri Toi River with associated pipeline and irrigation canals.

Environmental Assessment, Phase II LNG Terminal (PakGas), Port Qasim — *GEMS* | *Ecologist* | *From Mar 2017*
Assessed biodiversity sensitivities and risks of LNG handling at Port Qasim; discussed mitigation options with stakeholders.

Guidebook to Monitor Climate Change Finance — *CPNE (Council of Pakistan Newspaper Editors) for UNDP* | *Author* | *Nov 2016 – Jun 2017*
Wrote an accessible guidebook to help journalists monitor budgetary allocations for climate change.

Environmental & Social Assessment, Zephyr Wind Power Project, Gharo — *Tekcellent* | *Team Leader, Biodiversity* | *Sep 2016 – Nov 2016*
Assessed biodiversity sensitivities and risks during construction and operation of wind turbines; recommended mitigation.

Environmental Assessment, Korangi Thermal Power Station Phase II — *GEMS for K-Electric* | *Team Leader, Biodiversity* | *Jun 2016 – Nov 2016*
Assessed biodiversity risks of a hybrid LNG-operated turbine extension near Korangi Creek.

Environmental Assessment, Berth 3 & 4 Coal Transshipment (CPEC-linked) — *GEMS for Port Qasim Authority | Ecologist | Jun 2016 – Aug 2016*
Assessed biodiversity sensitivities and risks of coal-dust transshipment; discussed mitigation with stakeholders.

Plantation Plan, Karachi–Thatta Dual Carriageway — *EMC Pakistan for FWO | Lead Horticulturist | May 2016 – Jun 2017*
Developed a plantation plan spanning two ecological zones.

Environmental Assessment, Kadanwari Block — *SGS Pakistan for ENI | Team Leader, Biodiversity | Oct 2015 – Mar 2016*
Assessed biodiversity sensitivities and risks of petroleum exploration; recommended mitigation.

Jalalpur Irrigation Project EIA (ADB TA 8404-PAK) — *Asian Development Bank (through NESPAK-ICS JV and NEC Consultants) | Team Leader, Biodiversity | Aug 2015 – Oct 2015*
Biodiversity input for an irrigation project on the right bank of the Jhelum River, Khushab district.

ESMF & EIA, Integrated River Basin Management Project, Porali River — *The World Bank (through Techno Consult International) | Team Leader, Biodiversity | Jan 2014 – Mar 2014*
Biodiversity input for a basin-wide water, land and freshwater-ecosystem management project in Balochistan.

Environmental Assessment, Shah Bandar Block — *SGS Pakistan for Pakistan Petroleum Limited | Lead Consultant, Biodiversity | Jun 2013 – Mar 2014*
Assessed biodiversity sensitivities of petroleum exploration; study approved by the competent authority.

Project Management Workshop — *GEF-UNDP Small Grants Programme | Lead Facilitator | Nov 2013*
Guided applicants on submission protocols and helped the Secretariat present cases to the National Steering Committee.

Guinness World Record: Most Trees Planted in a Day — *Sindh Forest Department | Lead Independent Adjudicator | Jun 2013*
Adjudicated the record of 847,275 trees planted by a team of 300 at Kharochaan, Thatta, on 22 June 2013.

National Coordinator & Communication Focal Point, MFF – Pakistan — Apr 2009 – Jul 2012
Mangroves for the Future (MFF) programme, IUCN (International Union for Conservation of Nature) Pakistan

- Managed the MFF programme at national level in the top-level position, working with senior federal and provincial officials, CEOs of public and private organisations and heads of academic and research institutions.
- Led development and approval of the National Strategy and Action Plan (NSAP, Jul 2009 – Mar 2010) and the Guidelines for MFF Small Grants in Pakistan (Jan – Nov 2010).
- Convened seven National Coordinating Body (NCB) meetings with 85–90% member attendance; represented the NCB at the Regional Steering Committee with the country performance report.
- Managed the Small Grants Facility: convened expert review panels, contracted grantees and oversaw fund disbursements.
- Liaised with the regional secretariat and partners including UNDP, FAO and UNEP; briefed print and electronic media; delivered lectures on environmental and natural resource management.

Natural Resource Management (NRM) Coordinator — Jun 2005 – Mar 2009
IUCN Pakistan

- Provided NRM technical input to IUCN Pakistan programme and project offices nationwide and supported the Pakistan Ecosystem and Livelihoods Group (P-ELG) work plan.
- Member of the core team for Pakistan's National Wetlands Policy (draft, Jan 2007 – Jan 2009, for the Ministry of Environment, PWP-UNDP funded); coordinated stakeholders and led policy dialogue at sub-national and national levels.
- Supported proposal development, negotiation, inception and implementation of the ADB-funded Sindh Coastal Community Development Project; served as Ecologist (Mar 2008 – Jun 2012).
- Team Leader, Medicinal Flora of Ziarat (GEF-UNDP, Apr 2007 – Jun 2008), mainstreaming biodiversity into the production system of the Ziarat juniper forest; supported the GEF-UNDP Ziarat Juniper Project and BPSD NRM activities.
- Lead Ecologist, Biosaline Agriculture and Aquaculture Demonstration, Badin (Pakistan Environmental Programme, Sep 2005 – Aug 2007); Team Leader, Socioeconomic Baseline of Shigar Valley (Apr 2008 – Aug 2009).
- Country focal point for the IUCN Species Survival Commission, facilitating species and biodiversity initiatives; authored a sectoral paper on the flora of Sindh.

Consultant Environmental Specialist — Dec 1999 – Jan 2001
Environmental Assessment Services (EAS), IUCN Pakistan

- Provided technical support to the EAS unit for EIA and other impact assessment tools, including SEA.
- Conducted post-EIA evaluation by monitoring sulphur oxide emissions from a thermal power unit.

- Supported reassessment of a petroleum exploration company's practices and their impact on habitat and fauna near a protected area.

Biology / Botany Teaching and Departmental Leadership **Jun 1998 – Jun 2005**
Yaqeen Education Foundation; Aga Khan Education Services Pakistan & Aga Khan Higher Secondary School; Army Public College, Malir Cantt.

- Biology Teacher / Head of Science, Yaqeen Education Foundation.
- Lecturer / Head of Botany Department, Aga Khan Education Services Pakistan & Aga Khan Higher Secondary School.
- Lecturer / Head of Biology Department, Army Public College, Malir Cantt.

Project Manager, DGIS Mangrove Conservation Project **Aug 1997 – Dec 1997**
WWF-Pakistan

- Managed the establishment phase of a mangrove project funded by DGIS under the Tropical Rainforest Portfolio.

Consultant Conservation, Planning and Coordination Department **Oct 1995 – Dec 1996**
Pakistan State Oil Company (PSO)

- Coordinated with the Pakistan Environmental Protection Council and supported implementation of environment-related projects assigned to PSO.
- Negotiated with government agencies, managed procurement and BOQ documentation, called tenders and monitored bidding and site delivery.

Consultant Ecologist, Coastal Ecosystem Unit **Jan 1994 – Aug 1995**
IUCN Pakistan

- Contributed to monitoring & evaluation and feasibility of coastal ecosystem unit projects.
- Assessed survival and growth of mangrove species planted by the unit; conducted a feasibility study for an alternative apiculture site.

Research Fellow & Demonstrator **Feb 1992 – Apr 1993**
Plant Science Laboratories, University of Reading, UK

- Developed techniques and rapid HPLC/GLC analytical methods for isolating osmotica (quaternary ammonium compounds produced under osmotic stress).
- Assisted an ecophysiology study of coastal halophytes; demonstrated physiological ecology to undergraduate and postgraduate classes.

Research Officer & Cooperative Teacher, Department of Botany **Jan 1988 – Feb 1992**
University of Karachi

- Research officer on an NSRDB-funded project screening halophytes as food, feed and fibre resources along Pakistan's coastal deserts, mainly the Balochistan coast; the work contributed to the Institute of Halophyte Biology.
- Taught ecology to undergraduate and postgraduate classes.

EDUCATION

M.Sc. Botany (Plant Ecology), First Class — University of Karachi **1988**
B.Sc. (Honours) Botany, First Class — University of Karachi **1986**
Postgraduate Research Fellow — Plant Science Laboratories, University of Reading, UK **1992 – 1993**

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Postgraduate Certification in Environmental Health & Safety — PIQC / NED University, Karachi
- LEAD Pakistan Fellowship, Cohort 19 — focus on SDG 11, Urban Resilience (completed Jan 2017)
- International Training Workshop on Biosaline Agriculture Technology — PAEC with COMSTECH and HEC, NIAB Faisalabad (Feb 2006)
- Project Budgeting Programme for IUCN — LUMS Rausing Executive Development Centre, Lahore (Aug 2005)
- Monitoring & Evaluation Training Workshop — The Konsultant, Lahore (Aug 2005)
- National Training Course on Project Cycle Management (results-based approach & proposal writing) — MFF Secretariat, Karachi (Aug 2011)

PUBLICATIONS & TECHNICAL REPORTS

- Haq, R. (2016–2017). Guidebook to Monitor Climate Change Finance. CPNE for UNDP. (Author)
- Haq, R. (2010). Asian Plant Conservation Report — contributed country-level information in the development phase. (Contributor)
- Haq, R. (2009). Flora of Ziarat: Its Ethnobotanical and Medicinal Importance. IUCN / GEF-UNDP report.
- Haq, R. (2002). Flora of Sindh with its Ecological Significance. IUCN Sindh Programme project report.
- Haq, R. (1997). Rhizophora Plantation in Shahbunder: Survival and Growth Patterns. NORAD-IUCN joint project report.
- Haq, R. (1995). Rhizophora Plantation in Korangi and Phitti Creek: Survival and Growth Patterns. NORAD-IUCN joint project report.
- Haq, R. (1995). Wrong Lesson in Biodiversity. The News International, 8 December 1995.
- Haq, R. (1994). Vegetation Survey for Possible Apiculture in the Area Adjacent to the Indus Delta. UNDP-IUCN joint project report.
- Haq, R. & Khan, M.A. (1992). Screening of Halophytes as a Potential Source of Food. Proceedings of the National Conference on Problems and Resources of Makran Coast, Pakistan Council for Science and Technology, pp. 150–157.

PROFESSIONAL MEMBERSHIPS & ADVISORY ROLES

- Member, IUCN Commission on Ecosystem Management (CEM) — since Mar 2013
- Life Member, Horticultural Society of Pakistan — since Jan 1995; elected Executive Council member, 2015 – present
- Member, British Ecological Society — since Jan 1993
- Member, Pakistan Botanical Society — since Jan 1988
- Member, Advisory Group for DG Parks and Horticulture, KMC
- Member, Advisory Committee for Sindh EPA on technical issues
- Member, Core Group, Urban Forest Coalition

AWARDS & RECOGNITION

- Certificate of Appreciation — facilitating a World Environment Day 2010 session, Rotary Club of Karachi East (Jun 2010)
- Certificate of Appreciation — session on environmental issues, LEAP programme, Aga Khan University IED (Apr 2006)
- Certificate of Appreciation — supervising a prize-winning project at the provincial science exhibition, BIEK (Apr 2003)
- Certificates of Appreciation — conducting the First (Feb 1999) and Second (Feb 2000) International Courses on Wetland Management, University of Karachi
- Outstanding Social Leadership in the Field of Environment — Karachi Executive Lions Club (Oct 1995)

SELECTED LECTURES, WORKSHOPS & CONFERENCES

- Facilitator of two sessions, International Training Workshop on Conservation of Biodiversity and Ecosystems — IUCN CEM & Qatar University, Doha (Mar 2014)
- Lecture on Ecosystems and DRR, cohort training on Climate Change Adaptation and Disaster Risk Reduction — WWF & LEAD Pakistan, EU-funded (Sep 2014)
- Biodiversity Expert, Committee of Experts evaluating an EIA report — Sindh EPA (Jan 2014)
- Session contributor, IUCN CEM training workshops on Wetland Management, Biodiversity Conservation, and Environmental Assessment & Ecosystem Management — University of Karachi (2013)
- Participant, SAARC Workshop on Climate Change Impacts on Coastal & Aquatic Resources (Dec 2013); South Asian Environmental Assessment Conference, Islamabad (Dec 2013)
- MFF Regional Steering Committee meetings — Sri Lanka (2010) and Maldives (2011); MFF regional workshops on knowledge management & communications, Bangkok (2011, 2012)

LANGUAGES

English (fluent) • Urdu (fluent) *References available upon request.*